



Surface Pro 3 at a discount

Right on the heels of rumours about Microsoft's Surface Pro 4 tablet, the Surface Pro 3 is now available on sale



Xiaomi Mi TV 2

Xiaomi announced the launch of its affordable new 40-inch Android-powered smart TV for \$320
http://dgit.in/Xiao_TV

Geek life

Features	Oculus Rift	HTV Vive	Sony Morpheus	Samsung Gear VR	Razer OSVR
Resolution	1920 x 1080p (per eye)	1200 x 1080p (per eye)	960 x 1080p (per eye)	2560 x 1440 (combined)	1920 x 1080p (combined)
Display type	2x AMOLED displays	AMOLED display	LCD display	AMOLED display (requires phone)	AMOLED display
FOV	100° (Horizontal)	NA	90° (Horizontal)	96° (Horizontal)	90° H x 90° V
Display Refresh Rate	75 Hz	90 Hz	120 Hz	60 Hz	60 Hz
Sensors	Gyroscope, Accelerometer, Magnetometer	70 sensors for enhanced tracking	Cameras for tracking while moving	Accelerometer, Gyroscope, Geomagnetic, Proximity	Sensor hub with integrated accelerometer, Gyroscope and compass
Integrated Audio	Headphones	Headphones	Headphones	Headphones	Requires additional module

fledgling technology has seen a number of tech companies throw in their own VR solutions. The closest in terms of pure performance would have to be Razer's Open-Source Virtual Reality platform, aka, the Razer OSVR. Although the Oculus Rift is open source, Razer went one step further and made even the hardware open source — you can 3D print your

own dev kit as the files for it are available online.

Devs can use OSVR tools to develop platform-independent VR games and apps. Razer claims its main aim is to make VR more developer friendly. As a result, a number of video game studios and game engine developers, including big names such as Ubisoft, Untold Games, Virtuix, Unity 3D, and Unreal have pledged their support for the platform, ensuring a glut of potentially groundbreaking games to soon follow.

Microsoft is not one to simply watch the VR showdown from the sidelines. Their recently unveiled HoloLens integrates VR with augmented reality by allowing users to view and interact with holographic 3D projections set against a real-world envi-



Microsoft HoloLens



Razer OSVR

ronment. Featuring a lightweight, wireless headset that has a suspiciously uncanny resemblance to the RoboCop helmet, the HoloLens consists of a pair of lenses, a depth camera, a pair of speakers, and a translucent display used for the projected holograms (similar to what we've seen in Google Glass). The depth camera (in actuality, it is a group of cameras along the front and sides) is based on the same technology powering Microsoft's Kinect camera. It has a potential FOV of 120 degrees both horizontally and vertically and can track your movement as well as objects in the surrounding, including your hands, which makes potential applications endless. However, the project is in its nascent stage, and once a developer model is released, we can make a more informed judgement.

It looks like the contest for best VR headset will be a closely fought one, with Oculus, Sony, and HTC, all racing to finalize development of their own devices in order to capture those first-time sales, which are sure to be heavily influenced by this tech's novelty factor. Regardless, this year is set to be an exciting one for virtual reality gaming. And with such a prolonged period of development and hype, it seems more likely that the company that first manages to release a final consumer version will be the de facto winner. 

MIND OVER MATTER

While VR gaming is still struggling to achieve status as full-blown commercial tech, Switzerland-based neurotechnology company MindMaze has decided to go a few steps further with its thought-powered VR system dubbed "MindLeap." Developed originally for use in the medical field, the device consists of a head mounted display that comes equipped with motion-capture cameras coupled full 3D point cloud and skeleton extraction to track head and body movements. This enables users to experience a type of augmented reality, not unlike the system implemented by Microsoft's HoloLens. The device also uses sensors that detect brain and muscle activity allowing you (theoretically) to



control this augmented/virtual reality using just your thoughts. MindMaze hopes to release MindLeap's hardware engine and SDK in the coming months so game developers can use it to enhance the experience of other VR headsets.